TS-RAG: Retrieval Augmented Generation for Time Series Forecasting

Yixiong Xiao
xiaoyixiong@baidu.com
Baidu, Inc.
Beijing, China

Congxi Xiao
xiaocongxi@baidu.com
Baidu, Inc.
Beijing, China

Shuangli Li
Baidu, Inc.
Beijing, China

Jingbo Zhou*
jingbozhou@baidu.com
Baidu, Inc.
Beijing, China

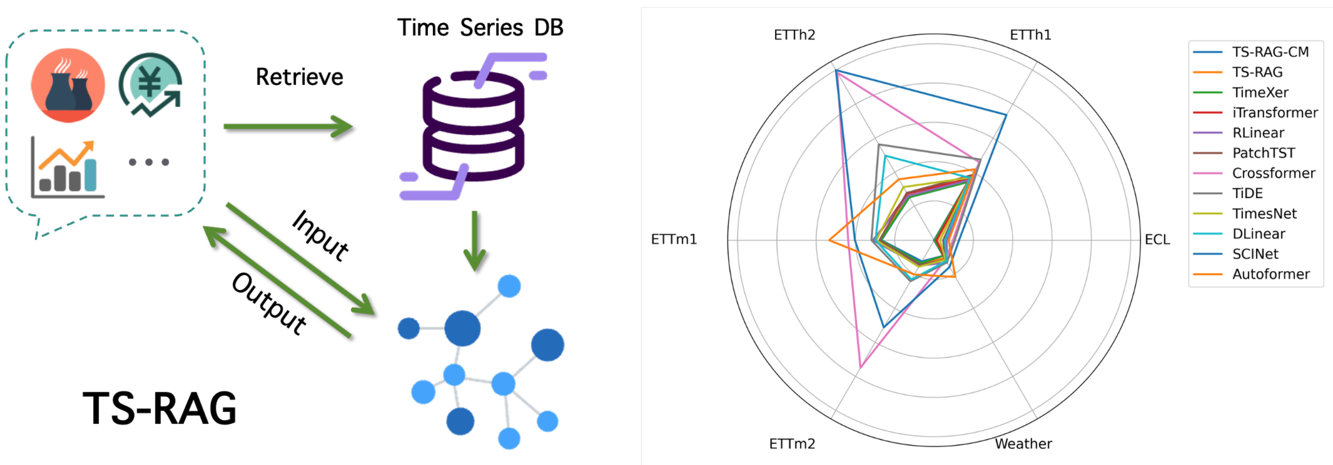

Figure 1: Left: schematic overview of TS-RAG. Right: model performance comparison on multiple forecasting benchmarks.

Abstract

While deep learning models, particularly transformer-based architectures, have shown impressive performance in time series forecasting, the application of retrieval-augmented generation (RAG) in this domain remains limited. Since RAG has proven effective in enhancing the capabilities of large language models by incorporating relevant external information, retrieving similar time series sequences as references might also improve accuracy in time series forecasting tasks. However, most time series models are constrained by limited training data, smaller parameter scales, and a lack of the extensive generative capabilities found in large language models. Simply concatenating reference sequences into the prompt, as done in language models, may not yield the expected results. To address these challenges, we propose a novel approach, TS-RAG, which leverages RAG to enhance forecasting performance. The framework introduces specially designed reference tokens to effectively fuse information from the input sequence with that from retrieved similar sequences, enabling a more robust capture of complex temporal dynamics. Experimental results demonstrate that TS-RAG achieves consistent state-of-the-art performance across several real-world forecasting benchmarks.

*Corresponding author.

Keywords

time series forecasting, retrieval augmented generation, RAG

1 Introduction

Time series forecasting plays a crucial role in numerous real-world applications, including finance, healthcare, energy, and climate science. The ability to accurately predict future values based on historical data is essential for decision-making processes across these domains. In recent years, deep learning models, particularly transformer-based architectures, have demonstrated remarkable performance in time series forecasting due to their ability to model long-range dependencies and capture complex temporal patterns [24]. These models excel at handling sequential data by using self-attention mechanisms that allow them to focus on relevant parts of the input sequence, even over long time horizons. Despite these advances, most existing models primarily focus on enhancing model training to improve accuracy, such as incorporating multi-scale mixing [33] and time imaging techniques [19]. These approaches aim to capture the intricate relationships and dependencies across different time scales within the time series. However, research into how to effectively leverage retrieval-augmented mechanisms to enhance forecasting accuracy is relatively scarce.